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Assistive devices non-use, abandonment, or non-adherence? Toward standard terminology for assistive devices outcomes

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ABSTRACT

For individuals with disabilities, failure to use prescribed assistive technology devices (ATDs) according to professional recommendations can have detrimental health consequences. The literature has employed various terms to describe this phenomenon such as nonuse, abandonment, and non-adherence to characterize this behavior, lacking clear and standardized definitions. Consistent use of a standardized language is critical for advancing research in this area. This study aims to identify and describe the concepts related to the failure to use prescribed ATDs, along with the associated contexts, and proposes a framework for standardizing terminology in this domain. A narrative literature review encompassing studies from inception to June 2023 was conducted to elucidate these concepts. Out of 1029 initially identified articles, 27 were retained for in-depth analysis. The review unveiled a significant inconsistency in the use of terms like nonuse, abandonment, noncompliance, and non-adherence. Some articles even employed these terms interchangeably without clear definitions. Only 10 of the 27 reviewed articles provided definitions for the terminology they used. This highlights the crucial need for adopting valid conceptual models to select appropriate terms. Researchers are strongly encouraged to furnish operational definitions aligned with theoretical models and relevant to their research context to advance this field consistently.

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KEYWORDS
assistive technology devices; discontinuance; non-compliance; non-use

Introduction

Assistive Technology Devices (ATD) is an umbrella term that includes systems and services related to assistive products such as, hearing aids, mobility assistive devices, communication aids, and memory aids, to mention a few (World Health Organization [WHO], 2018). In 2022, the International Organization for Standardization (ISO) provided an updated classification system for ATDs, aiding in their categorization and standardization. This system categorizes ATDs based on their functions, with each product assigned a unique code. These codes enable efficient communication between stakeholders and facilitate better accessibility to ATDs for individuals with disabilities (International Organization for Standardization, 2022). The classification encompasses a wide range of ATDs, including mobility aids, communication aids, assistive software, and more. By offering a standardized framework, ISO classification enhances collaboration among manufacturers, policymakers, healthcare professionals, and end-users, ultimately promoting the development and deployment of effective assistive solutions worldwide (Federici & Scherer, 2018).

Healthcare professionals often recommend ATDs to assist individuals with functional limitations to maintain or improve activity and participation (Carver et al., 2016; Salminen et al., 2009). There are various ways of obtaining ATDs. These devices can be acquired through two primary avenues: prescription by healthcare professionals or obtaining them "off-the-shelf" by users themselves (Gooberman-Hill & Ebrahim, 2007; Van der Esch et al., 2003). Healthcare professionals, such as physicians, occupational therapists, or physiotherapists, may prescribe ATDs based on an individual's specific needs and functional limitations. This process involves a thorough assessment of the person's condition, goals, and environmental factors to determine the most appropriate device. Prescribed devices are often customized or tailored to suit individual requirements, ensuring optimal fit and functionality. On the other hand, individuals may choose to obtain ATDs directly from the market without a formal prescription. These "off-the-shelf" devices are designed for general use and are readily available for purchase by users. While they may not offer the same level of customization as prescribed devices, they can still provide valuable support and assist individuals in managing their disabilities. The availability of both prescription-based and off-the-shelf options allows for flexibility in obtaining ATDs, catering to the diverse needs and preferences of users. However, non-prescribed ATDs can be obtained without any professional supervision which may hinder their optimal use (Khasnabis et al., 2010).

It is well-established that non-adherence (or failure to use ATDs as prescribed or recommended by healthcare professionals) can be detrimental to person's safety and is associated with poor quality of life (Liu, 2009; Luz et al., 2017; Stinchcombe et al., 2014) and associated with increased healthcare costs (Andrich et al., 2013). Recent studies reported that as many as 75% of clients are non-adherent to their ATDs (e.g.,

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Luz et al., 2017). The high prevalence of non-adherence and its negative consequences makes it a critical concern for researchers, clinicians, and others.

Multiple models have been developed over the past decades to describe the phenomena of ATDs use or nonuse by end-users. For example, the Matching Person and Technology (MPT) model offers valuable insights into the dynamics of ATD adoption by individuals with disabilities, shedding light on terms such as device nonuse, abandonment, and discontinuance (Scherer, 1996). According to this model, successful ATD adoption occurs when there is a harmonious alignment between the user's needs, abilities, and preferences, and the features and functions of the assistive device. Nonuse of assistive devices often arises when there is a mismatch between the user and the technology, either due to inadequate training, inappropriate device selection, or lack of customization (Scherer, 2002). Abandonment occurs when users discontinue the use of an ATD after initially accepting it, often due to dissatisfaction with its functionality, discomfort, or social stigma (Jutai et al., 2000). Discontinuance refers to the permanent cessation of ATD use, which can result from changes in the user's abilities, preferences, or circumstances. Understanding these terms through the lens of the MPT model highlights the importance of user-centered design, ongoing support, and customization in ensuring the successful integration of assistive technology into users' lives (Scherer et al., 2019).

Despite of the previous efforts, there is a noticeable lack of utilization of models and frameworks in defining the terms related to the non-usability of ATDs in the literature (Abrilahij & Boll, 2019). This lack of theoretical grounding can lead to inconsistent definitions and interpretations of terms. Without a clear conceptual framework, researchers may use these terms interchangeably or in a manner that does not align with the underlying factors influencing non-usability. For example, a study may refer to noncompliance when it actually pertains to nonuse, leading to confusion and difficulties in comparing findings across studies (Thomas et al., 2010). The limited use of models and frameworks hampers the advancement of research in this area, making it challenging to identify the specific reasons for non-usability and develop effective interventions to address the underlying barriers.

Consistent use of a standardized international language is critical for advancing research on ATDs outcomes (Elsaesser et al., 2022; Scherer et al., 2019); however, the existing research on ATDs includes a variety of terms for describing non-adherence. Many studies have used related terms such as noncompliance, nonuse, rejection, abandonment, and discontinuance but not necessarily in the same ways. Many researchers have failed to explain their choice of terminology in their studies and neglected to consider a clear method to measuring the term of their choice (Dahler et al., 2016; Sakakibara et al., 2013). This lack of uniformity in how non-adherence and its related concepts are defined and measured makes it difficult to draw meaningful comparisons between studies, even between those using the same terms.

Therefore, the aims of this research were to identify and describe concepts for failure to use prescribed ATDs, and the contexts in which they have been used, and propose a framework for standardizing language about these outcomes.

Method

In this study, a narrative literature review methodology was employed to address the predetermined objectives. Unlike systematic reviews, which have very narrowly defined parameters and precise eligibility criteria, a narrative review is more flexible in terms of the inclusion and exclusion rules (Allen, 2017). Narrative reviews are extremely valuable in synthesizing knowledge through summarizing and presenting similar concepts based on the published literature in a more abstract and meaningful way (Dixon-Woods et al., 2005). A narrative review also examines a given research question by including diverse empirical, theoretical, and methodological literature (Greenhalgh et al., 2018). In this work, there was no restriction on study designs, and thus, it was possible to present the findings based on different evidence types. A preliminary search of the literature determined that there has not been a review published on the topic of terminology for failure to use ATDs.

Search strategy

An online database search was performed on PubMed, CINAHL, Scopus, PsycINFO, and Web of Science. The database search strategy and selection of search terms was guided by a health sciences librarian to retrieve the most relevant citations from each database. All potentially relevant articles published in the English language from inception to June 2023 were included for further screening.
The search terms were a combination of the following sets:

Set 1: Any relevant term related to ATDs (e.g., assistive device, assistive technology, self-aid devices, rehabilitative devices, mobility assistive devices).

Set 2: Any relevant term related to non-usability (e.g., nonuse, non-adherence, noncompliance, abandonment, discontinuance, avoidance, rejection).

Within the sets, the sources were searched using OR relations, and between sets, AND relations were applied. For example, ("assistive device" OR "assistive technology" OR "self-aid devices" OR "rehabilitative devices" OR "mobility assistive devices") AND ("nonuse" OR "non-adherence" OR "noncompliance" OR "abandonment" OR "discontinuance" OR "avoidance" OR "rejection"). This search strategy resulted in the retrieval of 1016 articles, and an additional 13 articles were identified through other sources, resulting in 1029 articles that were eligible for screening.

Selection of articles for review

After removing the duplicates, screening of all the article titles and abstracts was performed using inclusion/exclusion criteria. To be included in this review, studies had to report research findings with any study design (i.e., quantitative, qualitative, or literature reviews) on any of the ATDs adoption outcomes outlined in the previous section. Articles were excluded if the full text was not available or if they were non-English publications. Also, editorials, correspondence papers, and opinion essays were

excluded from this review. Eventually, out of the 1029 articles identified for potential inclusion, 27 were retained for further analysis. Figure 1 presents the search strategy used in this review.

Data analysis

The data analysis process of this narrative review involved several key steps to synthesize and interpret the findings from multiple sources. Firstly, the first author identified relevant literature through comprehensive search strategies, including electronic databases, and reference lists. Once the articles were gathered, they were carefully screened for eligibility based on predetermined inclusion and exclusion criteria. Next, the validity of the selected articles was assessed to ensure the inclusion of reliable and relevant sources.

Following the screening stage, the first author, in collaboration with the second and third authors, extracted pertinent information from the selected articles, such as key terms, their definitions, study population and design, as well as whether devices were described by healthcare professionals. These extracted data were then organized and synthesized to identify patterns, trends, and commonalities across the literature. This synthesis process involved categorizing and grouping similar findings and terms to develop a coherent narrative framework. In the final stage, the researchers summarized the synthesized findings and presented them in a coherent narrative that aligns with the research objectives.

Results

In the Assistive Technology literature many terms are used interchangeably to describe the decision of not using ATDs. These terms include device *non-use* (de Boer et al., 2009; Edwards & Jones, 1998; Hass et al., 1996; Ishigami et al., 2021; Luz et al., 2017; Roelands et al., 2002; Skymne et al., 2012), *discontinuance* (Lauer et al., 2015; Petrie et al., 2018;

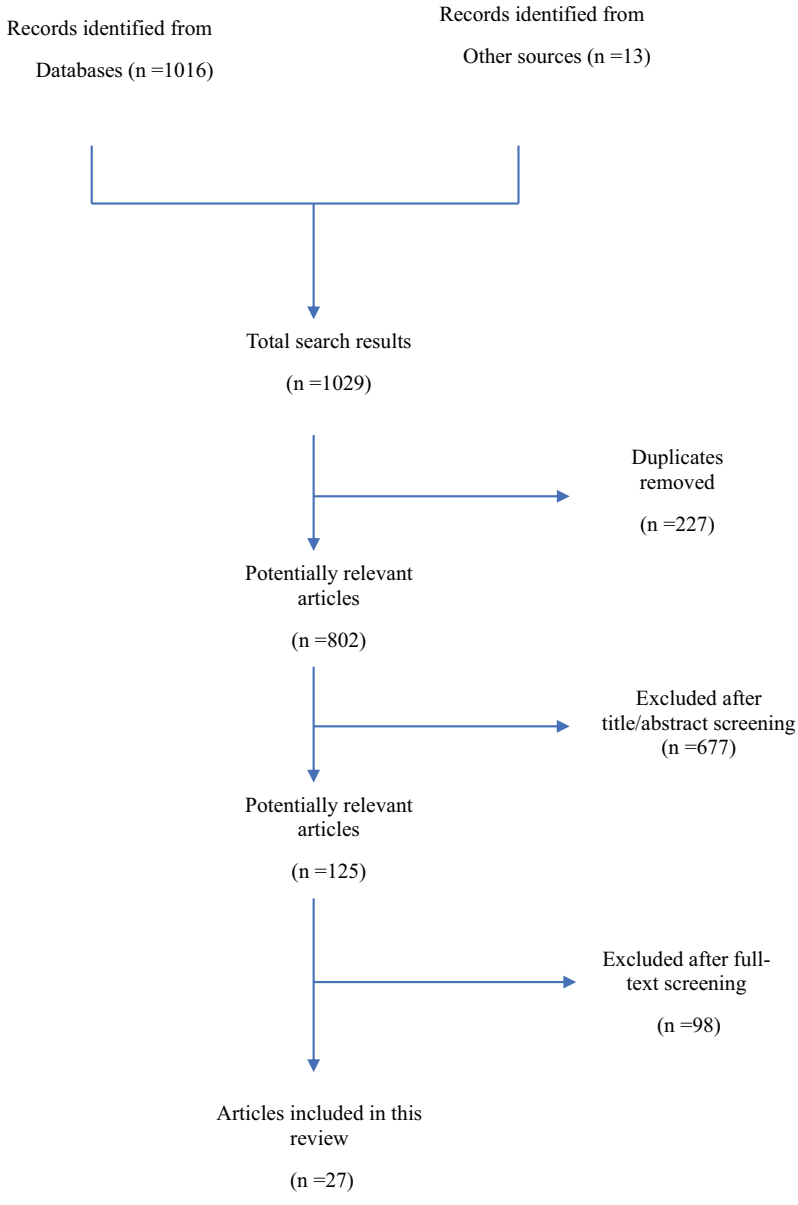


Figure 1. Search strategy.

Riemer-Reiss & Wacker, 2000), *abandonment* (Chen, 2020; Cruz et al., 2016; Federici & Borsci, 2016; Haggblom-Kronlöf & Sonn, 2007; Orellano-Colón et al., 2016; Phillips & Zhao, 1993; Sawadogo et al., 2022; Spiteri, 2018; Verza et al., 2006), *rejection* (Aminzadeh & Edwards, 1998), *disuse* (Gooberman-Hill & Ebrahim, 2007; Yeh, 2009), *non-acceptance* (Barker et al., 2004; Hedberg-Kristensson et al., 2007), or *avoidance* (Lilja et al., 2003). All of these terms are discussed in the following section.

The existing literature encompasses various terminologies and definitions, highlighting the complexity of this phenomenon. Out of the 27 reviewed studies, only 10 provided operational definitions of the terms they used. One commonly used term is "abandonment," defined as using the device on less than 50% of occasions on which the related activity was performed (Sawadogo et al., 2022). The same term was defined slightly different by Phillips and Zhao as an absolute nonuse of a device at the time of the survey. Verza et al. (2006) defined abandonment as the disuse of a previously obtained device for any reason.

Furthermore, there seems to be no differentiation among the studies in selecting and using terms as a function of whether ATDs was prescribed. For example, Edwards and Jones (1998) included a sample of community dwellings to examine the ownership and use of several forms of ATDs. They did not ask about who provided the devices to study participants. Federici and Borsci (2016) examined device abandonment and included only individuals with prescribed or recommended ATDs. Phillips and Zhao (1993) used the same term (i.e., abandonment) and included a sample from the community with prescribed and non-prescribed ATDs. Similarly, Thomas et al. (2010) examined ATDs noncompliance among a sample of older adults who underwent total hip replacement surgery and included those who purchased their devices on their own and those who had their devices prescribed by healthcare professionals.

In the literature, there has been limited research that investigated the concept of non-adherence to the use of ATDs. The literature review identified only one study (literature review) that specifically looked at *non-adherence* (Tuazon et al., 2018). The rest of the reviewed studies used other terms (see Table 1). A concise and unified definitions of these terms may help healthcare professionals and researchers better distinguish between the action of not using a device (e.g., nonuse or abandonment) and the action of not using the device in the context of rehabilitation plans (e.g., noncompliance or non-adherence).

The term "non-use" has been used widely in the literature to refer to the scenario when a device is not used (Abrilahij & Boll, 2019; Chen et al., 2000; Dijcks et al., 2006). The most straightforward way to define nonuse is to answer the following question: "Is the device being used at this moment in time, yes or no?" (Dijcks et al., 2006). The term nonuse seems to be a neutral description of the phenomenon.

Some authors have used the term nonuse in their studies, where in fact they were looking at non-adherence. Examples of these studies include those by Luz et al. (2017) and Skymne et al. (2012), which included participants with prescribed ATDs and in which the psychosocial reasons of nonuse were investigated. This inappropriate application of the term nonuse by some authors may be, as mentioned by Wessels et al. (2003), due to the lack of a solid theoretical framework that provides the definition and explains the mechanism of device nonuse.

Among the terms mentioned above, the term "abandonment" is the most common in literature and refers to a somewhat active or conscious decision to stop using a device (Wessels et al., 2003). In this sense, abandonment is a more absolute term that often indicates that the device is no longer used at all and may represent only a small percentage of cases (Swinnen & Kerckhofs, 2015).

Some authors have intentionally used the term abandonment to refer to a decision resulting from a motivational emotional reaction. Individuals may sometimes be dissatisfied, or they decide that the device is not a good fit for their lifestyle or environment. However, other researchers inappropriately use the term abandonment, and thus interpretations of their findings can be misleading.

One example is the study by Phillips and Zhao (1993), who employed a definition of abandonment that described only one outcome of a person-device interaction (use vs. nonuse), and they considered a device to be abandoned if it was replaced by another device. In this case, the authors were talking about situations where a person felt that they needed more functional support, and so, they may have stopped using a cane in favor of using a walking frame, for instance. Thus, the authors used the word abandonment in this context to imply that an individual stopped using one type of device in favor of another, which is different from other researchers' use of the term abandonment. Also, Phillips and Zhao's definition of abandonment ignores the changes from one category of devices to another due to dissatisfaction with the original device. The authors did, however, acknowledge this limitation, and they clearly stated that "'expected' abandonment was not distinguished from abandonment due to user dissatisfaction" (Phillips & Zhao, 1993).

Meanwhile, Chen (2020) used a slightly different definition of abandonment than Phillips and Zhao (1993). Chen's definition included emotional reasons for device abandonment. However, this study measured device abandonment in the exact same way, i.e., by looking at the use or nonuse of a device. Interestingly, Chen (2020) concluded that a user's dissatisfaction or frustration could lead to device abandonment.

Federici and Borsci (2016) investigated device abandonment in a sample of older adults in four healthcare centers in Italy who had obtained their devices under professional supervision. Their definition of abandonment was similar to that of Chen's (2020), and measured it in the same way, i.e., by examining device use/nonuse. They also provided motivational reasons for device abandonment. They concluded that users' frustration or dissatisfaction with their devices and with the service delivery and the follow-up processes were the main reasons for device abandonment. However, the context of Federici and Borsci's (2016) study and the way they discussed their findings were different from those of Phillips and Zhao (1993) and Chen (2020). Federici and Borsci's (2016) often referred to the relationships between prescribers and the end-

users and emphasized a person-centered approach in assistive device recommendation. This definition is more relevant for device non-adherence or noncompliance, but not for device abandonment.

Table 1 presents a selection of articles that used only one term consistently and provided a reasonably clear definition or at least a description of the situation in which the term used. Only 10 articles provided an operational definition of the used terms (Chen, 2020; Federici & Borsci, 2016; Ishigami et al., 2021; Lauer et al., 2015; Luz et al., 2017; Phillips & Zhao, 1993; Sawadogo et al., 2022; Thomas et al., 2010; Tuazon et al., 2018; Verza et al., 2006). Articles that used more than one term interchangeably were excluded from the table for clarity purposes. Table 1 presents the different terms from the most neutral terms, such as nonuse, to the more specific ones, such as noncompliance and non-adherence. The articles were also arranged in chronological order with reference to whether the authors defined the term and if they included participants with prescribed devices. The latter is important because it elucidates the study context.

Discussion

This narrative literature review aimed to provide an in-depth understanding of the different terms used by researchers to describe situations of failure to use ATDs, and the contexts in which they have been used. The findings of this review suggest a large discrepancy among researchers in using terms such as nonuse, discontinuance, abandonment, noncompliance, and non-adherence that needs further attention to advance the field further. The following discussion provides an overview of the current use of these terms with reference to their theoretical and clinical implications.

The discrepancy in using inconsistent terms carries significant implications. These terms, despite being related to the same overarching concept of not using a device as prescribed or intended, can lead to confusion and misinterpretation among professionals, researchers, and users alike. Some of the terms mentioned above lack specificity and clarity and can thus be confusing. Most of these terms refer to the negative impression of making the decision (i.e., the person consciously

Table 1. Terms associated with nonuse that were identified in the literature.

				In what context(s) did the authors use the term?			
#	Author(s) (citation)	Defining term	Definition cited	Study population	Study design	MAD was prescribed?	Did the study examine reasons for non-use?
1	Ishigami et al. (2021)	Non-use	Yes	Home-dwellings[a] Canada	Cross-sectional	No	No
2	Luz et al. (2017)	Non-use	Yes	Home-dwellings USA	Cross-sectional	Yes	Yes
3	Skymne et al. (2012)	Non-use	No	Community-dwellings[b] Sweden	Focus groups	Yes	Yes
4	de Boer et al. (2009)	Non-use	No	Community-dwellings Netherlands	Cross-sectional	Yes	Yes
5	Roelands et al. (2002)	Non-use	No	Home-dwellings Belgium	Cross-sectional	No	Yes
6	Edwards and Jones (1998)	Non-use	No	Community-dwellings UK	Cross-sectional	No	No
7	Hass et al. (1996)	Non-use	No	Home-dwellings Sweden	Qualitative interviews	Yes	No
8	Lauer et al. (2015)	Discontinuance	Yes	Community-dwellings USA	Cross-sectional	No	Yes
9	Riemer-Reiss and Wacker (2000)	Discontinuance	No	Home-dwellings USA	Cross-sectional	Yes	Yes
10	Sawadogo et al. (2022)	Abandonment	Yes	Home-dwellings France	Observational	Yes	No
11	Chen (2020)	Abandonment	Yes	Community-dwellings Hong Kong	Qualitative interviews	No	Yes
12	Spiteri (2018)	Abandonment	No	Community-dwellings Malta	Qualitative interviews	Yes	Yes
13	Orellano-Colón et al. (2016)	Abandonment	No	Community-dwellings Puerto Rico	Qualitative interviews	No	Yes
14	Cruz et al. (2016)	Abandonment	No	Home-dwellings Brazil	Cross-sectional	No	Yes
15	Federici and Borsci (2016)	Abandonment	Yes	Community-dwellings Italy	Cross-sectional	Yes	Yes
16	Haggblom-Kronlöf and Sonn (2007)	Abandonment	No	Home-dwellings Sweden	Cross-sectional	No	Yes
17	Phillips and Zhao (1993)	Abandonment	Yes	Home-dwellings USA	Cross-sectional	No	Yes
18	Verza et al. (2006)	Abandonment	Yes	Home-dwellings Italy	Observational	Yes	Yes
19	Aminzadeh and Edwards (1998)	Rejection	No	Community-dwellings Canada	Focus groups	No	Yes
20	Yeh (2009)	Disuse		Community-dwellings Home-dwellings Australia	Mixed methods	No	Yes
21	Gooberman-Hill and Ebrahim (2007)	Disuse	No	Home-dwellings UK	Qualitative interviews	No	Yes
22	Hedberg-Kristensson et al. (2007)	Non-acceptance	No	Home-dwellings Sweden	Focus groups	Yes	Yes
23	Barker et al. (2004)	Non-acceptance	No	Home-dwellings Canada	Qualitative interviews	Yes	Yes
24	Lilja et al. (2003)	Avoidance	No	Home-dwellings Sweden	Qualitative interviews	No	Yes
25	Thomas et al. (2010)	Non-compliance	Yes	Hospitalized[c] USA	Qualitative interviews	Yes	Yes
26	Gitlin et al. (1998)	Non-compliance	No	Community-dwellings USA	Structured interview	Yes	Yes
27	Tuazon et al. (2018)	Non-adherence	Yes	Community-dwellings Home-dwellings Hospitalized	Literature review	Yes	Yes

[a]Home-dwellings = individuals are residing in their own home.
[b]Community-dwellings = individuals are residing in long-term care units, assisted-living homes, nursing homes, or senior housing.
[c]Hospitalized = inpatients due to health conditions that need close medical attention.

decides not to use the device). However, some terms such as "discontinuance" may connote a valid reason for not using a device (e.g., healing, updating the device, or even deterioration of the functional status) and thus are helpful. Without unified terminologies to a given context, comparing research findings, identifying patterns, and developing evidence-based interventions or policies becomes challenging.

Methodologically, the measurement of these terms can vary, ranging from self-report single question about use or nonuse to more detailed investigations of patterns of use, duration of use, frequency of use, and intention to use a device in future. These differences in terminology and methodologies highlight the need for clarity and consensus to facilitate accurate measurement, comparisons, and interventions targeting ATDs utilization. As evidenced in this study, there seems to be inconsistent measurement of these terms across studies. To clarify, device nonuse generally refers to situations where individuals do not use the prescribed device at all (Luz et al., 2017). Abandonment typically implies a more permanent and deliberate decision to cease using the device, often accompanied by an intention not to resume its use (Sawadogo et al., 2022; Verza et al., 2006; Wessels et al., 2003). Discontinuance signifies a temporary interruption in device use, with the possibility of resuming its use at a later time (Fuhrer et al., 2003; Riemer-Reiss & Wacker, 2000). Noncompliance and non-adherence are terms commonly used in the clinical context and typically involve a broader scope of psychosocial and behavioral factors, including not only reasons for device nonuse, but also reasons for inconsistent or improper use (Tuazon et al., 2018).

A point of note is that non-users are not always non-adherent or non-compliant. For example, the functional needs of individuals may change so that they no longer need to use the device. Likewise, a client may use their prescribed device only once a month, and so while they can technically be labeled "device users," in fact the client is non-adherent or non-compliant because they do not follow the professional recommendation, which is usually to use the device daily (or at least three times per week) (Evans et al., 2007; Wressle & Samuelsson, 2004). This simple example reflects the complexity and sensitivity of picking one term over another because there are methodological implications for each (as discussed above). A term's concise definition and the population it applies to, in addition to the variables and the outcome measures regarding the manner and times of usage, are all methodological considerations that should be in account to provide consistent findings.

To better understand the nuances of non-usability terms, it is helpful to look at some conceptual models that explain this phenomenon.

Rogers (1983), in his model, explained the behavior of device use and discontinuance and introduced two important components of this process namely, the adoption decision and the diffusion process. According to Roger's philosophy, the process by which end-users transition through from the adoption of an innovation to its diffusion consists of a five-stage decision-making process. These stages include knowledge, persuasion, decision, implementation, and confirmation (Rogers, 1983, 2003).

Knowledge and persuasion phases take place before the adoption decision, as the potential user explores the available options of devices, becomes aware of the device of interest, and collects persuasive information about its features. This followed by the conscious decision to adopt or reject the device. If the adoption is considered, the potential user moves to the implementation phase, in which the prescription process and training of use takes place. Finally, the end-users evaluate whether the device meets their needs and expectations. If the needs and expectations are met, the user moves to the state of continued use. If the user is unsatisfied with the device, the discontinuance decision may be taken (Riemer-Reiss, 1999; See Figure 2).

The Roger's process view of adoption and diffusion of innovations is helpful in conceptualizing the different phases of ATDs use from initiation phase all the way to the discontinuance decision. As documented by several authors, adopting such generic process view is helpful to capture both the static and the dynamic nature of device discontinuance phenomenon (Beer et al., 2017; Riemer-Reiss & Wacker, 2000; Soliman & Rinta-Kahila, 2020; Van De Ven, 1992). Discontinuance, however, can happen in any of the five stages mentioned above, as noted by Rogers (2003).

Fuhrer et al. (2003) in their framework for the conceptual modeling of ATD outcomes described two types of discontinuance: namely, shorter-term and longer-term discontinuance. The shorter-term discontinuance takes place in the introductory phase, while the longer-term discontinuance can happen anytime after the introductory period. Both types of discontinuance can happen as a result of an interaction between device use and a number of moderating cofactors that include personal and environmental factors, among others (Fuhrer et al., 2003). The term discontinuance has a neutral connotation, referring to the termination of device use after a period of time.

Discontinuance is the most objective way to describe whether a person has continued or stopped using a device. Discontinuance in the classic scenario is a decision to stop using a device after previously accepting it. According to Rogers' theory of diffusion (Rogers, 2003), there are two reasons for an individual's decision to discontinue using a device: replacement and disenchantment. The *replacement* (e.g., stop using a device for an updated one) and *disenchantment* (e.g., stop using a device due to dissatisfaction). The replacement decision is a valid reason for discontinuance that may happen due to improvements or deterioration of functional capacities of clients. However, disenchantment decision is the unfavored discontinuance that concerns all the stakeholders of ATDs services.

As mentioned above, this literature review also identified other terms, such as *rejection*, *disuse*, *avoidance*, *non-acceptance*, and *abandonment*. As stated by Lauer et al. (2015), "these terms connote negative meanings, when in fact the reason for not using a device may be related to a positive factor for an individual (e.g., healing)."

Despite the progress brought by previous research, it is valid to say that the efforts in the Assistive Technology literature are fragmented due to a lack of conceptual uniformity when using different terms that apply to different situations. It

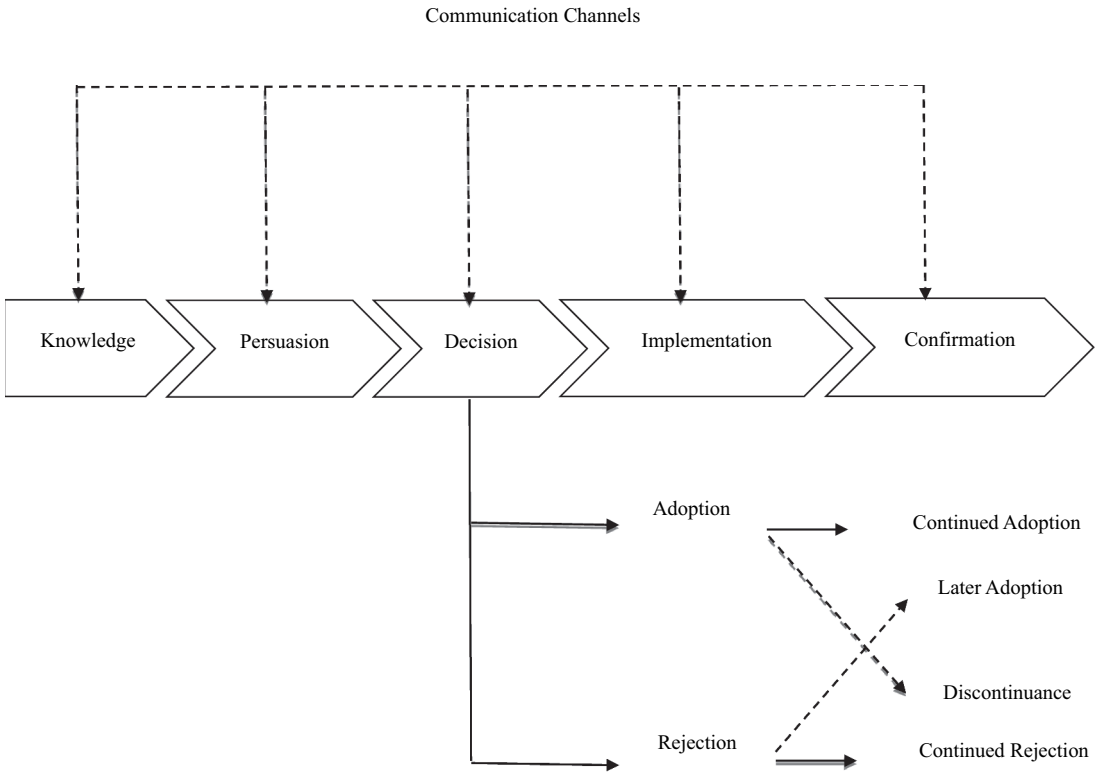


Figure 2. Rogers' model.

is also difficult to draw meaningful comparisons between studies, even those using the same terms. The term abandonment is often misleading, for example, especially when applying it when clients stop using their devices due to improvement or deterioration in their physical functioning (Verza et al., 2006). Therefore, some authors have called for replacement of the term "abandonment" with the more neutral term "discontinuance" because the term abandonment suggests a negative impression that is not representative of all situations where devices are no longer being used (Lauer et al., 2015).

Non-adherence, on the other hand, is a clinically oriented concept and is a special case of discontinuance. The term non-adherence is applicable when a device is prescribed or recommended by qualified healthcare professionals within a rehabilitation plan. Thus, the reasons for non-adherence fall into the disenchantment category of Rogers' model (Rogers, 1983). That is, the client stopped using the prescribed device, not because of improvement or deterioration of their functional capacities, but due to psychosocial reasons.

Based on this definition, non-adherence implies a more active and conscious decision to not use the prescribed device. This definition also places great emphasis on the client's role in making informed decisions among the available device options. In this sense, more collaborative involvement of the client and more work with their clinicians is needed in planning and implementing ATDs.

In clinical practice, healthcare professionals are constantly giving instructions to their clients on how to use their prescribed devices to meet a predetermined healthcare goal. Clients, however, can either follow the instructions or ignore them. When clients do not follow their instructions, they fall under one of two concepts, called "non-adherence" or "non-compliance," which respectively refer to different points of view. To better understand the concept of ATDs non-adherence, therefore, it is helpful to make a conceptual distinction between non-adherence and noncompliance, as the two concepts are commonly confused.

The term noncompliance first appeared in the medical literature in the 1970s, and since then has been used to describe a quantifiable parameter of "the extent to which a patient's behavior (in terms of taking medications, following diets or executing other lifestyle changes) coincides with the clinical prescription" (Jay et al., 1984). In this classic situation, the prescriber informs the client how to use the treatment regimen. Thus, the role of clients' opinions is neglected (Vrijens et al., 2012). In recent years, however, noncompliance has been criticized for the negative implication that clients are expected to passively follow clinicians' orders (Ahmed & Aslani, 2014; Hugtenburg et al., 2013).

As stated by Wessels et al. (2003), "(Non-) compliance relates to how well the patient obeys (or complies with) the doctor's orders. The patient is perceived as a passive receiver of instructions. Non-adherence, on the other hand, relates to the willingness and the possibility of a patient to develop his own health plan. Adherence has to do with commitment. The patient is perceived as someone who has the right to make choices, guided by the professional."

The main challenge that comes up from this review lies in the long list of concepts used by researchers to refer to similar yet different observations without clear and consistent definitions. Examples of these concepts include nonuse, non-

acceptance, rejection, disuse, abandonment, noncompliance, and non-adherence. This is a valid concern because such inconsistency in the use of these different terms may hinder the development of the field. There is no clear chronological order of the appearance of these terms in the literature; however, it seems that the field of Assistive Technology adopted different definitions over time. Although there have been some attempts to explain and differentiate between these terms, there is still a need to clearly define and distinguish between them.

In line with this concern, the developers of the Consortium for Assistive Technology Outcomes Research (CATOR) framework acknowledged that some confusion exists (Jutai et al., 2005). Therefore, they encouraged researchers to be clear about what their core phenomenon and what goes around it. In other words, what is the core concept that has the most neutral meaning, and what are the other special cases of that core concept? There is a simple observable fact of whether or not somebody is using a device. In this context, terms like "continuance" and "discontinuance" may be relevant; people either continue to use the device or they do not. This simple way is easily verifiable and not too controversial. Then, there are terms that go further and imply motivations or reasons for using or not using a device, like "abandonment" or "misuse." Finally, the terms "non-compliance" and "non-adherence" are more clinically oriented concepts and refer to special cases that involve healthcare professionals, caregivers, or social circles.

Another relevant point in this discussion is whether or not device prescription was involved in the adoption process. For instance, if someone simply bought an assistive device on their own initiative and stopped using it, it has nothing to do with noncompliance or non-adherence because the device was not prescribed by a professional. As well, if a device was not prescribed by a qualified healthcare professional, then there is no negotiated healthcare plan to adhere to. Thus, the terms noncompliance or non-adherence are not relevant.

Clear distinctions should be made between terms like non-compliance and non-adherence, which involve a relationship between prescriber professional and users, terms that imply a reason why somebody stopped using a device like abandonment or rejection, and neutral terms such as nonuse and discontinuance. In the current literature, however, this distinction is not an easy task because not all authors use these terms in a common, consistent way. Some authors say that devices are sometimes deliberately rejected or abandoned because of a reason (e.g., emotional), as in Chen's (2020) study, while others use these terms just because they had been used before and they just did some surveys around and used the term abandonment while documenting whether people were using the device or not (e.g., Phillips and Zhao (1993)).

How non-adherence-related terms and concepts are defined in the current literature makes it difficult to draw meaningful comparisons between studies. This inconsistency may be partly due to the complexity of studying the non-adherence phenomenon (Jahan et al., 2024). However, it is essential for researchers and clinicians to consider specific definitions of their terms and to provide a clear context for their intended measures to help maintain the continuum of the Assistive Technology domain. We echo the recommendations from Scherer et al. (2019) that this process should embrace both evidence-based practice and practice-based evidence perspectives.

The proposed framework for selection of terms

From our literature review and conceptual analysis, we propose a framework for selecting appropriate ATD outcomes. The framework aims to provide clarity and consistency in terminology usage while considering the various stages and factors involved in device adoption, use, and discontinuance. The framework is structured based on the stages outlined in Rogers' model of adoption and diffusion of innovations, the CATOR framework, as well as the distinctions made between different terms used in the literature. The proposed framework includes the following:

(1) Core Concepts:
   - Continuance: Refers to the ongoing use of an assistive device by the user.
   - Discontinuance: Indicates the cessation of device use by the user for any reason.

(2) Special Cases of Discontinuance:
   - Non-Adherence: Occurs when an individual takes an active choice of not to use the prescribed device as recommended by healthcare professionals, primarily due to psychosocial reasons. In this case, the client is perceived as someone who has the right to make choices, guided by the professional.
   - Non-Compliance: Describes the extent to which a user obeys or complies with the clinical prescription provided by healthcare professionals regarding device use in a passive manner without active involvement in decision-making.

(3) Reasons for Discontinuance:
   - Replacement: Occurs when a user stops using a device to transition to an updated or alternative device.
   - Disenchantment: Refers to discontinuance due to user dissatisfaction with the device, stemming from various factors such as functionality, usability, or psychosocial aspects.

(4) Neutral Terms:
   - Non-Use: Denotes the absence of device use without implying any specific reason or judgment.
   - Disuse: Similar to nonuse, but may imply a temporary cessation of device use.

(5) Considerations:
   - Prescription Involvement: It is crucial to consider whether the device was prescribed by a qualified healthcare professional. Terms like non-adherence and noncompliance are relevant only in the context of prescribed devices with a healthcare plan.
   - User Involvement: Recognize the importance of user involvement in decision-making regarding device adoption, use, and discontinuance. Terms like abandonment or rejection may imply user-driven decisions.
   - Clear Definitions: Ensure clear and consistent definitions of terms used in research and clinical practice

to facilitate meaningful comparisons between studies and maintain conceptual clarity within the field.

This framework provides a structured approach for selecting appropriate assistive device outcomes. By considering core concepts, special cases, reasons for discontinuance, and neutral terms, researchers and clinicians can effectively characterize device use dynamics and facilitate evidence-based decision-making in ATD interventions. This framework may potentially contribute to the advancement of the field and the improvement of ATDs services.

Implications of the proposed framework

The proposed framework for selecting appropriate ATD terms presents significant implications across various domains. It enhances research standardization by providing clear definitions and distinctions between terms, thereby facilitating comparability of study findings and advancing knowledge in assistive technology research. In clinical practice, healthcare professionals can utilize the framework to assess device usage patterns, identify reasons for discontinuance, and tailor interventions, promoting evidence-based decision-making and improved service delivery. By emphasizing a user-centered approach, the framework has the potential to promote user satisfaction, engagement, and overall quality of life. Additionally, policymakers can leverage the framework to inform policy development and resource allocation strategies, ultimately enhancing ATDs provision and utilization.

Study limitations

This study has a few limitations that should be acknowledged. The first limitation may be due to the nature of the narrative review methodology. Since the narrative review method is arguably not systematic, it is important to consider this limitation when interpreting the findings. Although efforts were made to thoroughly search the relevant literature, it is possible that a few articles were missed due to methodological limitations. For example, due to the long list of terms used in the literature to refer to non-adherence, the search strategy might have failed to capture some studies that used less common terms. Also, since only literature in English was included in this review, results from studies that were published in other languages are not represented.

Future research directions

While the proposed framework provides a valuable foundation for understanding non-adherence and related terms, there remain several avenues for future research to explore. Specifically, future research may consider an exploratory methodology, such as using a grounded theory lens to identify terminology use across contexts. Additionally, longitudinal studies examining the trajectory of non-usability over time, as well as the effectiveness of interventions aimed at promoting adherence, are needed to better understand the dynamic nature of device usage behaviors. By continuing to explore these areas and building upon the proposed framework, researchers can contribute to ongoing advancements in the field and ultimately improve outcomes for individuals relying on assistive devices.

Conclusion

This review of the current literature has sought to unify the inconsistent terminology under the guidance of existing models and frameworks that provides a potentially helpful roadmap for future research. Based on the findings of this review, it is valid to say that there is inconsistency in the use of terms such as abandonment, nonuse, noncompliance, and non-adherence in the literature on assistive technology. Also, the term abandonment seems problematic, as researchers have used it widely to investigate slightly different situations without clear definitions. In this current study, an attempt was made to cut through some of that confusion, and to accomplish this, a conceptual distinction was made between the various terms, and operational definitions were provided based on published theories and models (e.g., the CATOR framework and Rogers' theory of diffusion). Finally, this work has uncovered crucial literature gaps and methodological biases in the current literature that future researchers may consider improving the quality of their findings. To help advance the field, the recommendation for researchers is that they should provide conceptual and operational definitions of terms related to nonuse and apply these terms consistently in their work.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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